

Note : Whenever the sampling is with replacement, the population is considered infinite.

For example, consider a box with 5 balls in it. If, when a sample is drawn, the balls are replaced each time they are drawn, a sample of size $n = 10$ or $n = 100$ or whatever size is desired can be drawn. Hence the population is considered infinite.

f.p.c is necessary in theory but seldom necessary in practice because the sampling fraction $\frac{n}{N}$ is usually less than 0.05 which makes f.p.c close to one.

Property 3. When f.p.c approaches one the standard error of mean is simplified as

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

• or the variance mean as

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \quad \text{or} \quad v(\bar{x}) = \frac{\sigma^2}{n}$$

Example 14.10.1

For a population 0, 4, 8, 12 construct sampling distribution of mean for samples of size 2 taken without replacement and find its mean and standard error.

Solution :

When sampling is done without replacement, all possible samples = ${}^N C_n = {}^4 C_2 = 6$

Since every sample is equally likely to occur, the probability of each sample is $\frac{1}{6}$.

The probability of each sample mean is therefore also $\frac{1}{6}$.

Sample No.	All possible samples	Sample mean $\bar{x}$	P ($\bar{x}$)
1	0, 4	2	1/6
2	0, 8	4	1/6
3	0, 12	6	1/6
4	4, 8	6	1/6
5	4, 12	8	1/6
6	8, 12	10	1/6

The sampling distribution of $\bar{x}$ is tabulated below :

$\bar{x}$	P ($\bar{x}$)
2	1/6
4	1/6
6	2/6
8	1/6
10	1/6

Mean $\mu_{\bar{x}}$ and standard error $\sigma_{\bar{x}}$ of this distribution are computed as

$\bar{x}$	$P(\bar{x})$	$\bar{x} \cdot P(\bar{x})$	$\bar{x}^2 \cdot P(\bar{x})$
2	1/6	2/6	4/6
4	1/6	4/6	16/6
6	2/6	12/6	72/6
8	1/6	8/6	64/6
10	1/6	10/6	100/6
	1	$\frac{36}{6} = 6$	$\frac{256}{6}$

$$\mu_{\bar{x}} = \sum \bar{x} \cdot P(\bar{x}) = \frac{36}{6} = 6$$

$$\sigma_{\bar{x}}^2 = \sum \bar{x}^2 \cdot P(\bar{x}) - [\sum \bar{x} \cdot P(\bar{x})]^2$$

$$= \frac{256}{6} - (6)^2$$

$$= \frac{40}{6} = \frac{20}{3}$$

$$\sigma_{\bar{x}} = \sqrt{\frac{20}{3}}$$

These results can be obtained using population mean μ and standard deviation σ .

Population mean and standard deviation are computed as

x	x^2
0	0
4	16
8	64
12	144
24	224

$$\mu = \frac{\sum x}{N} = \frac{24}{4} = 6$$

$$\sigma^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2$$

$$= \frac{224}{4} - (6)^2$$

$$= 20$$

$$\text{or } \sigma = \sqrt{20}$$

$$\text{Now } \mu_{\bar{x}} = \mu = 6$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{\sqrt{20}}{\sqrt{2}} \sqrt{\frac{4-2}{4-1}}$$

$$= \sqrt{\frac{20}{3}}$$

Example 14.10.2

For a population 0, 4, 8, 12, construct the sampling distribution of mean for samples of size 2 taken with replacement and find its mean and standard error.

Solution :

When samples are drawn with replacement, all possible samples = $(N)^n = (4)^2 = 16$

Since 16 samples are random, every sample is equally likely to occur having probability $\frac{1}{16}$. Thus every sample mean has probability $\frac{1}{16}$.

Possible samples	Sample mean ($\bar{x}$)	P ($\bar{x}$)
0, 0	0	1/16
0, 4	2	1/16
0, 8	4	1/16
0, 12	6	1/16
4, 0	2	1/16
4, 4	4	1/16
4, 8	6	1/16
4, 12	8	1/16
8, 0	4	1/16
8, 4	6	1/16
8, 8	8	1/16
8, 12	10	1/16
12, 0	6	1/16
12, 4	8	1/16
12, 8	10	1/16
12, 12	12	1/16

Now the sampling distribution of $\bar{x}$ is

$\bar{x}$	P ($\bar{x}$)
0	1/16
2	2/16
4	3/16
6	4/16
8	3/16
10	2/16
12	1/16
Total	1

Computation of mean and standard error.

$\bar{x}$	P ($\bar{x}$)	$\bar{x} \cdot p (\bar{x})$	$\bar{x} \cdot P (\bar{x})$
0	1/16	0	0
2	2/16	4/16	8/16
4	3/16	12/16	48/16
6	4/16	24/16	144/16
8	3/16	24/16	192/16
10	2/16	20/16	200/16
12	1/16	12/16	144/16
Σ	1	96/16	736/16

- (i) Calculate the performance which batsman is more consistent
(05 MARKS)
(ii) Calculate mean deviation about median for batsman A
(iii) Calculate the skewness for batsman A and B and compare the result
(iv) Calculate Standard deviation for batsman B
(v) Calculate coefficient deviation for batsman B

Batsman (A)	12	15	0	76	48	4
Batsman (B)	47	12	76	48	4	1

STATISTICS

Sampling Distribution Theory

$$\begin{aligned}\mu_{\bar{x}} &= \sum \bar{x} \cdot P(\bar{x}) \\ &= \frac{96}{16} = 6 \\ \mu_{\bar{x}} &= 6 \\ \sigma_{\bar{x}}^2 &= \sum \bar{x}^2 \cdot P(\bar{x}) - (\sum \bar{x} \cdot P(\bar{x}))^2 \\ &= \frac{736}{16} - \left(\frac{96}{16}\right)^2 \\ &= 46 - 36 = 10 \\ \sigma_{\bar{x}} &= \sqrt{10}\end{aligned}$$

These results can be obtained directly as

$$\begin{aligned}\mu_{\bar{x}} &= \mu = 6 \\ \sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{20}}{\sqrt{2}} = \sqrt{10}\end{aligned}$$

14.11 STANDARD ERROR OF MEAN

The standard error of mean, $\sigma_{\bar{x}}$, is always less than the population standard deviation σ . It depends on the size of the sample drawn. If the sample size increases, the standard error of mean decreases and consequently the value of sample mean will be closer to the value of population mean. This phenomenon is described in another way : as the standard error decreases the precision with which the sample mean can be used to estimate population mean increases.

A random sample of size 6 is drawn without replacement from a finite population of size 16. The standard deviation of the population is 12.

- Find standard error of mean.
- Find standard error of mean if, n is increased to 10.

Solution :

- Standard error of mean for a finite population when the sampling is without replacement is given by the formula.

$$\begin{aligned}\sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \\ N &= 16 \\ n &= 6 \\ \sigma &= 12 \\ \sigma_{\bar{x}} &= \frac{12}{\sqrt{6}} \sqrt{\frac{16-6}{16-1}} = 4\end{aligned}$$

- (ii) The sample size is increased to 10

Now

$$N = 16$$

$$n = 10$$

$$\sigma = 12$$

$$\sigma_{\bar{x}} = \frac{12}{\sqrt{10}} \sqrt{\frac{16-10}{16-1}} = 2.4$$

Example 14.11.2

A random sample of size 16 is drawn from a population with replacement. The standard deviation of the population is 5.

- Find standard error of mean.
- Find standard error of mean if n is increased to 100.

Solution :

- When sampling is with replacement, the f.p.c is ignored and standard error of mean is written as

$$\begin{aligned}\sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{n}} \quad ; \quad \sigma = 5, n = 16 \\ &= \frac{5}{\sqrt{16}} = \frac{5}{4}\end{aligned}$$

- For $\sigma = 5$

$$n = 100$$

$$\begin{aligned}\sigma_{\bar{x}} &= \frac{5}{\sqrt{100}} = \frac{1}{2} \\ &= 0.5\end{aligned}$$

Example 14.11.3

A random sample of size 25 is drawn from a population with mean 82 and standard deviation 30. It is desired to reduce the standard error of mean by $\frac{1}{3}$. What size sample must be selected to accomplish this?

Solution :

Population size is not given so we have to assume it large (or infinite). Therefore

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad (\text{f.p.c. is ignored})$$

$$\text{and } n = 25$$

$$\sigma = 30$$

(i) Calculate the performance which batsman is more consistent					
(ii) Calculate mean deviation about median for batsman A					
(iii) Calculate the skewness for batsman A and B and compare the result					
(iv) Calculate Standard deviation for batsman B					
(v) Calculate coefficient deviation for batsman B					
Batsman (A)	12	15	8	73	7
Batsman (B)	47	12	76	48	4

STATISTICS

Sampling Distribution Theory

$$\sigma_{\bar{x}} = \frac{30}{\sqrt{25}}$$

$$\sigma_{\bar{x}} = 6$$

The standard error, $\sigma_{\bar{x}} = 6$ is to be reduced to $\sigma_{\bar{x}} = 2$ (one third)

Now

$$\sigma = 30$$

$$\sigma_{\bar{x}} = 2$$

Sample size n can be found using the relation

$$n = \frac{\sigma^2}{\sigma_{\bar{x}}^2}$$

$$\text{Thus } n = \frac{(30)^2}{(2)^2} = 225$$

14.12 CENTRAL LIMIT THEOREM : SHAPE OF THE SAMPLING DISTRIBUTION OF $\bar{x}$

So far the mean and standard error of the sampling distribution of mean have been discussed but not the shape of distribution. An important theorem known as central limit theorem describes the shape of the sampling distribution of mean.

The theorem states :

The sampling distribution of mean is normal distribution either if the population is normal or if the sample size is more than 30

Thus

- (i) If the population is normal, the sampling distribution of mean must be normal irrespective of the size of sample.
- (ii) If the population is not normal, the sampling distribution of mean is approximated by a normal distribution provided that the sample size is large.

A sample of size greater than 30 is considered large.

For $\bar{x}$ is a normally distributed variable with mean μ and standard error

$$\frac{\sigma}{\sqrt{n}} \cdot Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ is a standard normal variable with mean zero and standard error one.}$$

To clarify further, the central limit theorem may be presented as :